Supplementary proofs to "A Fast Elementary Series to Calculate Limb-darkened Transit Flux"

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This supplement contains the proofs that are not needed to use or to check the method of the paper: the error bounds for the truncation of the interior series by powers of p (Section S1), and the proof of the bound for the contact pieces, Theorem 3 of the paper (Section S2). Equation numbers with a prefix S refer to this supplement. Readable explanations of all proofs, and notebooks that check every bound numerically against references independent of the derivations, are in the same repository (`docs/proofs` and `validation`).

Notation from the paper

The stellar disk is the unit disk, r the distance from its centre, $\mu = \sqrt{1-r^2}$, and $D_\gamma(z,p) = \langle (1-r^2)^\gamma \rangle$ the mean of $(1-r^2)^\gamma$ over an occulting disk of radius p centred at distance z; $w = 1-z^2$, $v = 1/w$, $\beta_m = (-1)^m \binom{\gamma}{m}$. Truncated by powers of p, the series of the paper is $S_N^{\mathrm{P}} = \sum_{n\le N} T_n$, where T_n is the coefficient of p^{2n}, equal to the n-th term of the mean-value expansion

$$\langle f \rangle = \sum_{n=0}^{\infty} \frac{(p^2/4)^n}{n!\,(n+1)!}\, \Delta^n f\big|_{\mathrm{centre}}, \tag{S1}$$

with Δ the Laplacian in the sky plane. Across the limb and inside the disk, the blocked integral $P_\gamma = \int (1-r^2)^\gamma \,\mathrm{d}A$ over the occulted part of the star is, with $s = r^2$, $a = (z-p)^2$, $b = (z+p)^2$, $s_1 = 1$ across the limb and $s_1 = b$ inside,

$$P_\gamma = -\frac{1}{2(\gamma+1)} \int_a^{s_1} (1-s)^{\gamma+1} \left[1 + \frac{p^2-z^2}{s}\right] \frac{\mathrm{d}s}{\sqrt{(s-a)(b-s)}}, \tag{S2}$$

With $x = 1-(z-p)^2$, $c_0 = 1+p^2-z^2$, $c_j = (p^2-z^2)x^j$ $(j \ge 1)$,

$$k = \frac{1-(z-p)^2}{4zp}, \tag{S3}$$

and $\kappa = 1/k$,

$$P_\gamma = -\frac{x^{\gamma+1}\sqrt{k}}{2(\gamma+1)} \sum_{j\ge 0} c_j\, B\left(\tfrac{1}{2}, \gamma+2+j\right)\, {}_2F_1\left(\tfrac{1}{2}, \tfrac{1}{2}; \gamma+\tfrac{5}{2}+j; k\right), \tag{S4}$$

$$P_\gamma = -\frac{\pi\, x^{\gamma+1}}{2(\gamma+1)} \sum_{j\ge 0} c_j\, {}_2F_1\left(-\gamma-1-j, \tfrac{1}{2}; 1; \kappa\right). \tag{S5}$$

For fixed p, z is an algebraic function of k,

$$z = p(1-2k) + \sqrt{p^2(1-2k)^2 + 1 - p^2}, \tag{S6}$$

Theorem 1 (Theorem 3 of the paper) *Let $p < \frac{1}{2}$. On every piece of the contact expansions, $|P_\gamma - \hat{P}_\gamma| \le E_\gamma^{(j)} + E_\gamma^{(n)}$, where $\hat{P}_\gamma$ is the polynomial approximation of the piece, $E_\gamma^{(j)}$ (Equation S15) bounds the remainder of the j-series and $E_\gamma^{(n)}$ (Equations S20–S21) the truncation in n. Both are explicit functions of p, γ, J, N, the piece and a radius ρ, and are finite whenever the disk conditions of Lemma 2 hold. The flux error is at most $\sum_i |c_i|\,(E_{\gamma_i}^{(j)} + E_{\gamma_i}^{(n)}) \big/ \big(2\pi \sum_i c_i/(\alpha_i + 2)\big)$.*

S1 Truncation by powers of p

Truncation by powers of p gives a second family of approximations, $S_N^{\mathrm{P}} = \sum_{n \le N} T_n$, which is also a polynomial in v, of degree $2N$. Two bounds apply to it.

Proposition 1 (Mean-value remainder) *Let $f \in C^{2N+2}$ on the closed occulting disk $\bar{D}$. Then*

$$\langle f \rangle - \sum_{n=0}^{N} \frac{(p^2/4)^n}{n!(n+1)!} \Delta^n f\big|_{\mathrm{centre}} = \frac{(p^2/4)^{N+1}}{(N+1)!\,(N+2)!}\, \Delta^{N+1} f(\xi) \tag{S7}$$

for some $\xi \in \bar{D}$. For $f = (1 - r^2)^\gamma$ with $0 < \gamma < 1$, $\Delta^n f \le 0$ and $|\Delta^n f|$ increases with r for $n \ge 1$. The remainder therefore lies between the right-hand side evaluated at $r = \max(0, z - p)$ and at $r = z + p$.

The proof, which follows Pizzetti (1909), is given in Section S1.1. The upper end of this two-sided bound grows like $(p/(1 - z - p))^{2N+2}$ and is useful only for $z < 1 - 2p$. A bound valid up to contact follows from the sign structure.

Proposition 2 (Abel bound) *Let $\gamma = \frac{1}{2}$, $t = p^2$ and $t_1 = (1 - z)^2$. Then for $0 \le p \le 1 - z$,*

$$0 \le S_N^{\mathrm{P}}(z, p) - D_{1/2}(z, p) \le \left(\frac{p}{1 - z}\right)^{2N+2} \left[S_N^{\mathrm{P}}(z, 1 - z) - D_{1/2}(z, 1 - z)\right], \tag{S8}$$

where the internally tangent disk has the elementary mean

$$D_{1/2}(1 - q, q) = \frac{2 \arccos(1 - 2q) - \frac{4}{3}(3 + 2q - 8q^2)\sqrt{q(1 - q)}}{3\pi q^2}, \qquad 0 < q < 1. \tag{S9}$$

Proof. As a power series in t, $D_{1/2}(z, \sqrt{t}) = \sum_n d_n t^n$ with $d_n \le 0$ for $n \ge 1$, convergent for $t < t_1$. The partial sums at t_1 are non-increasing in N and bounded below by $D_{1/2}(z, 1 - z)$, because $S_N^{\mathrm{P}}(z, \sqrt{t}) \ge D_{1/2}(z, \sqrt{t})$ for $t < t_1$ and $D_{1/2}$ is continuous in p. The series therefore converges at t_1, and by Abel's theorem its sum is $D_{1/2}(z, 1 - z)$. Then $\sum_{n>N} |d_n| t^n \le (t/t_1)^{N+1} \sum_{n>N} |d_n| t_1^n$. Equation (S9) is the $b + r = 1$ case of Agol et al. (2020), divided by q^2; it agrees with direct quadrature to 10^{-27}. □

S1.1 Mean-value expansion with remainder on the disk

Let $f \in C^{2N+2}(\bar{D})$, where $\bar{D}$ is the closed disk of radius p about c, and let $m_g(\rho)$ denote the mean of g over the circle of radius ρ about c. Green's theorem gives

$$2\pi\rho\, m_f'(\rho) = \int_{|x-c| \le \rho} \Delta f \, \mathrm{d}A = 2\pi \int_0^\rho s\, m_{\Delta f}(s)\, \mathrm{d}s. \tag{S10}$$

Integrating from 0 gives

$$m_f(\rho) = f(c) + (\mathcal{T} m_{\Delta f})(\rho), \qquad (\mathcal{T} g)(\rho) = \int_0^\rho s \ln \frac{\rho}{s}\, g(s)\, \mathrm{d}s. \tag{S11}$$

The operator $\mathcal{T}$ has a non-negative kernel and maps ρ^{2j} to $\rho^{2j+2}/(2j+2)^2$. Iterating, with $m_{\Delta^j f}(0) = \Delta^j f(c)$,

$$m_f(\rho) = \sum_{j=0}^{N} \frac{\rho^{2j}}{4^j (j!)^2} \Delta^j f(c) + (\mathcal{T}^{N+1} m_{\Delta^{N+1} f})(\rho). \tag{S12}$$

The area mean is $\langle f \rangle = (2/p^2) \int_0^p \rho\, m_f(\rho)\, \mathrm{d}\rho$. The polynomial terms give the coefficients $(p^2/4)^j/(j!(j+1)!)$ of Equation (S1). The remainder is $\int_0^p K(s)\, m_{\Delta^{N+1} f}(s)\, \mathrm{d}s$ with a non-negative kernel K, whose integral equals the remainder for $\Delta^{N+1} f \equiv 1$, namely $(p^2/4)^{N+1}/((N+1)!(N+2)!)$. Since each circle mean lies between the minimum and maximum of $\Delta^{N+1} f$ on $\bar{D}$, the mean-value theorem for integrals gives Equation (S7).

For $f = (1-r^2)^\gamma = \sum_m \beta_m r^{2m}$ with $0 < \gamma < 1$, $\Delta^n r^{2m} = 4^n \left(m!/(m-n)!\right)^2 r^{2(m-n)}$ for $m \geq n$. Hence $\Delta^n f = \sum_{m \geq n} \beta_m 4^n (m!/(m-n)!)^2 r^{2(m-n)}$ has only negative coefficients for $n \geq 1$. It is therefore negative, and its modulus increases with r. On the disk r ranges over $[\max(0, z-p), z+p]$ (for $z < p$ the disk covers the stellar centre), which gives the two-sided bound of Proposition 1.

S2 Proof of the bound for the contact pieces

This section proves Theorem 1 of the paper (restated above). Throughout, $0 < p < \frac{1}{2}$, $m = \gamma + \frac{3}{2}$, and u is the variable of a piece: $u = k$ (first contact), $u = y = 1 - k$ and $u = y' = 1 - \kappa$ (second contact), $u = k - k_0$ and $u = \kappa - \kappa_0$ (middle pieces centred at k_0 and κ_0). On each piece the variable satisfies $|u| \leq w$. For a power series $F(u) = \sum_n t_n u^n$ we write $\hat{F}(r) = \sum_n |t_n| r^n$ and $\hat{F}^{>n_0}(r) = \sum_{n > n_0} |t_n| r^n$.

S2.1 Representation

In Equation (S2) substitute $s = 1 - x(1-t)$. Then $s - a = xt$, $b - s = 4zp(1-kt)$ because $b - 1 + x = b - a = 4zp$, and $\mathrm{d}s = x\, \mathrm{d}t$, so that

$$P_\gamma = -\frac{x^{\gamma+1}\sqrt{k}}{2(\gamma+1)} \int_0^{t_1} t^{-1/2} (1-t)^{\gamma+1} (1-kt)^{-1/2} \left[1 + \frac{p^2 - z^2}{s}\right] \mathrm{d}t, \tag{S13}$$

with $t_1 = 1$ across the limb and $t_1 = 4zp/x = \kappa$ inside the disk. For $z > p$ we have $0 < x < 1$ and $0 \leq x(1-t) \leq x$, so $1/s = \sum_j x^j (1-t)^j$ converges uniformly on $[0,1]$ and may be integrated term by term. With

$$H_j(k) = \int_0^1 t^{-1/2} (1-t)^{\gamma+1+j} (1-kt)^{-1/2} \mathrm{d}t, \qquad G_j(\kappa) = \frac{1}{\pi} \int_0^1 t^{-1/2} (1-t)^{-1/2} (1-\kappa t)^{\gamma+1+j} \mathrm{d}t \tag{S14}$$

(the second after $t \to \kappa t$), Euler's integral (Olver et al., 2010, 15.6.1) gives $H_j = B(\frac{1}{2}, \gamma + 2 + j)\, {}_2F_1(\frac{1}{2}, \frac{1}{2}; \gamma + \frac{5}{2} + j; k)$ and $G_j = {}_2F_1(-\gamma - 1 - j, \frac{1}{2}; 1; \kappa)$, and hence Equations (S4) and (S5). We write $\mathcal{F}_j = H_j$, $\Pi = -x^{\gamma+1}\sqrt{k}/(2(\gamma+1))$ across the limb and $\mathcal{F}_j = \pi G_j$, $\Pi = -x^{\gamma+1}/(2(\gamma+1))$ inside, so that $P_\gamma = \Pi \sum_j c_j \mathcal{F}_j$, and $P_{\gamma,J}$ for the sum through $j = J$.

S2.2 The remainder in j

Lemma 1 *Let a piece have $z_- \le z \le z_+$ with $z_- > p$ and $k \le \bar{k}$ (limb). With $\bar{x} = 1-(z_- - p)^2$ and $\bar{Q} = z_+^2 - p^2$,*

$$|P_\gamma - P_{\gamma,J}| \le E_\gamma^{(j)} = \frac{\bar{x}^{\gamma+1}\bar{Q}\,\bar{x}^{J+1}}{2(\gamma+1)(1-\bar{x})} \times \begin{cases} \sqrt{\bar{k}}\,H_{J+1}(\bar{k}) & \text{across the limb,} \\ \pi & \text{inside the disk.} \end{cases} \tag{S15}$$

Proof. For $j > J$, $|c_j| = |z^2 - p^2|\,x^j$. The integrands of Equation (S14) are positive, and they do not increase with j because $0 \le 1-t \le 1$ and $0 \le 1-\kappa t \le 1$ for $0 \le \kappa \le 1$. Hence $0 < H_j \le H_{J+1}$ and $0 < \pi G_j \le \pi G_{J+1} \le \pi$ for $j > J$, and $|P_\gamma - P_{\gamma,J}| \le |\Pi|\,|z^2-p^2|\,\mathcal{F}_{J+1}\sum_{j>J} x^j = |\Pi|\,|z^2-p^2|\,\mathcal{F}_{J+1}x^{J+1}/(1-x)$. On the piece, x decreases and z^2-p^2 increases with $z > p$, k decreases with z across the limb, and H_{J+1} and $\sqrt{k}$ increase with k. Each factor is therefore bounded by its value at the appropriate end of the piece. For $\bar{k} = 1$, $H_{J+1}(1)$ follows from Gauss's sum (Olver et al., 2010, 15.4.20); for $\bar{k} < 1$ it equals $\hat{H}_{J+1}(\bar{k})$, which Lemma 4 bounds. □

S2.3 Analytic continuation of the geometry

Lemma 2 *(a) Across the limb, $z = p(1-2k) + W^{1/2}$ with $W = 1 - 4p^2k(1-k)$. On a closed disk D on which $4p^2|k|\,|1-k| < 1$, the principal root $W^{1/2}$ is analytic and agrees with Equation (S6) on $D \cap \mathbb{R}$, z has no zero, and there are branches of $(4pz)^{\gamma+1}$, of $x^{\gamma+1} = (4pzk)^{\gamma+1}$ and of $\sqrt{k}$ (the latter two if $0 \notin D$) that are analytic on D, agree with the real ones on $D \cap \mathbb{R}$, and satisfy $|f^{\gamma+1}| = |f|^{\gamma+1}$. (b) Inside the disk, $z = \big(V^{1/2} - p(2-\kappa)\big)/\kappa$ with $V = (\kappa - 2p^2)^2 + h^2$, $h^2 = 4p^2(1-p^2)$. If $D = \{|\kappa - \kappa_0| \le r\}$, κ_0 real, does not contain 0 and*

$$\min_{-1 \le v \le 1} \left[c^2 + h^2 - r^2 + 2crv + 2r^2v^2\right] > 0, \qquad c = \kappa_0 - 2p^2, \tag{S16}$$

the same conclusions hold with $x = 4pz/\kappa$.

Proof. (a) $|W-1| < 1$ implies $\operatorname{Re} W > 0$, so the principal root is analytic on D; it is positive on $D \cap \mathbb{R}$, as in Equation (S6). Since $\big(W^{1/2} + p(1-2k)\big)\big(W^{1/2} - p(1-2k)\big) = W - p^2(1-2k)^2 = 1 - p^2 \ne 0$, z has no zero on D. A function without zeros on a disk has an analytic logarithm, unique up to $2\pi i$, which we fix by its real value on $D \cap \mathbb{R}$; then $|f^{\gamma+1}| = |\exp((\gamma+1)\log f)| = |f|^{\gamma+1}$. (b) Solving $\kappa = 4zp/(1-(z-p)^2)$ for z gives the stated root. $\operatorname{Re} V$ is harmonic, so its minimum over D lies on the circle $\kappa = \kappa_0 + re^{i\theta}$, where $\operatorname{Re} V$ is the bracket of Equation (S16) with $v = \cos\theta$. Thus $\operatorname{Re} V > 0$ on D, and the principal root is analytic and positive on $D \cap \mathbb{R}$. From $\big(V^{1/2} - p(2-\kappa)\big)\big(V^{1/2} + p(2-\kappa)\big) = (1-p^2)\kappa^2$, a zero of $V^{1/2} - p(2-\kappa)$ requires $V = p^2(2-\kappa)^2$ and hence $\kappa = 0 \notin D$. The rest follows as in (a). □

On D the crude bounds $|z| \le p(1 + 2\max_D|k|) + (1 + 4p^2 \max_D|k| \max_D|1-k|)^{1/2}$ (limb) and $|z| \le \big(d + r + p(2 + \max_D|\kappa|)\big)/\min_D|\kappa|$ (inside, with $d = |\kappa_0 - 2p^2 - ih|$, since $|V| \le (d+r)^2$) hold, together with $|x| \le 4p \max_D|z| \max_D|k|$ and $|x| \le 4p \max_D|z|/\min_D|\kappa|$.

Lemma 3 *Let f be analytic on $|u| \le \rho_2$ with $|f| \le C$ there, and $\rho < \rho_2$. Then*

$$\max_{|u|=\rho} |f(u)| \le \max_{0 \le \ell < K} \big|f(\rho e^{2\pi i \ell/K})\big| + \frac{\pi\rho\,C}{K(\rho_2 - \rho)}. \tag{S17}$$

Proof. For $|u| = \rho$, Cauchy's estimate on the disk of radius $\rho_2 - \rho$ about u gives $|f'(u)| \le C/(\rho_2 - \rho)$. Every point of the circle lies within arc length $\pi\rho/K$ of a sample, and $|f|$ changes along the arc by at most the arc length times $\max|f'|$. □

We apply Lemma 3 with $K = 1024$ to z and x on a disk of radius $\rho_2 > \rho$ that satisfies Lemma 2, with the crude bounds as C, and call the results Z and X. With $Q = p^2 + Z^2$, on $|u| = \rho$

$$|c_0| \le C_0 = 1 + Q, \qquad |c_j| \le C_j = Q\,X^j \quad (j \ge 1). \tag{S18}$$

S2.4 Majorants of the Gauss functions

Lemma 4 *(a) Let $t_{n+1}/t_n = (n+a)(n+b)/((n+c)(n+1))$ and let $n+a$, $n+b$, $n+c > 0$. Then for all $n' \ge n$*

$$\left|\frac{t_{n'+1}}{t_{n'}}\right| \le \bar r(n) = 1 + \frac{\max(0, a+b-c-1)}{n+c} + \frac{|ab-c|}{(n+c)(n+1)}, \tag{S19}$$

and if $q = \bar r(n)\, r < 1$, then $\sum_{n' \ge n} |t_{n'}| r^{n'} \le |t_n| r^n/(1-q)$. (b) For $x \ge \frac16$, $\ln x - 1/x \le \psi(x) \le \ln x$. (c) If $|1-\kappa| \le 1$ and $\kappa \notin [1,\infty)$, then $|G_j(\kappa)| \le 1$. (d) For $0 < w < 1$, $\max_{0<y\le w} y^m |\ln y| = L^*(w)$, with $L^*(w) = w^m |\ln w|$ for $w \le e^{-1/m}$ and $1/(em)$ otherwise.

Proof. (a) $(n+a)(n+b) - (n+c)(n+1) = n(a+b-c-1) + ab - c$ and $n \le n+1$ give the bound for $n' = n$; $\bar r$ does not increase with n. The tail is then dominated by a geometric series. (b) Binet's formula (Olver et al., 2010, 5.9.13) gives $\psi(x) = \ln x - 1/(2x) - 2\int_0^\infty t\,\mathrm{d}t/\big((t^2+x^2)(e^{2\pi t}-1)\big)$. The integral is positive and at most $x^{-2}\int_0^\infty t\,\mathrm{d}t/(e^{2\pi t}-1) = 1/(24x^2)$, so $\ln x - 1/(2x) - 1/(12x^2) \le \psi(x) \le \ln x$, and $1/(12x^2) \le 1/(2x)$ for $x \ge \frac16$. (c) On $t \in [0,1]$, $1 - \kappa t$ lies on the segment from 1 to $1-\kappa$, so $|1-\kappa t| \le 1$ and $1 - \kappa t \ne 0$; the weight in Equation (S14) integrates to 1. (d) $y^m|\ln y|$ increases on $(0, e^{-1/m}]$ and decreases on $[e^{-1/m}, 1)$. □

The series about $k = 0$ have $a = b = \frac12$, $c = \gamma + \frac52 + j$, and all their coefficients are positive, so $|H_j(k)| \le \hat H_j(|k|)$. About argument 1 the connection formulae (Olver et al., 2010, 15.8.4, 15.8.10) write $\mathcal{F}_j = A_j(u) + u^{m+j}\big[B_{l,j}(u)\ln u + B_{c,j}(u)\big]$ with $u = y$ or y'. For non-integer m, A_j and $u^{-m-j}(\mathcal{F}_j - A_j)$ are again hypergeometric series, with parameters $(a, b; 1-m-j)$ and $(c-a, c-b; 1+m+j)$, and $B_{l,j} = 0$. For integer m (half-integer γ), A_j is a polynomial, $B_{l,j}$ is hypergeometric with parameters $(a+m+j, b+m+j; 1+m+j)$, and $B_{c,j}$ has the coefficients of $B_{l,j}$ multiplied by $\psi_n = \psi(a+m+j+n) + \psi(b+m+j+n) - \psi(1+n) - \psi(1+m+j+n)$. All arguments of ψ here are at least $\frac12$. By Lemma 4(b) and $|\ln(X/Y)| \le |X-Y|/\min(X,Y)$, $|\psi_n| \le \Phi(n)$, where $\Phi(n)$ is built from terms $\alpha/(n+\beta)$ with $\alpha \ge 0$, $\beta > 0$ by sums and a maximum, and so does not increase with n. The tail of $\hat B_{c,j}$ after the nth term is therefore at most $\Phi(n+1)$ times that of $\hat B_{l,j}$. With Lemma 4(a), every $\hat F(r)$ and $\hat F^{>n_0}(r)$ is bounded by finitely many terms and a geometric tail.

S2.5 Truncation in n

Lemma 5 *Let $f = \tilde f + e$ with f analytic on $|u| \le \rho$ and $|f| \le M$ on $|u| = \rho$, let $\tilde f_n$ and e_n be the Taylor coefficients of $\tilde f$ and e, and let $\sum_{n \le N_B} |e_n| w^n \le \eta'$. Then for $|u| \le w < \rho$ and $N \le N_B$,*

$$\left|f(u) - \sum_{n\le N} \tilde f_n u^n\right| \le \sum_{n=N+1}^{N_B} |\tilde f_n|\, w^n + M\,\frac{(w/\rho)^{N_B+1}}{1 - w/\rho} + \eta'. \tag{S20}$$

Proof. $f(u) - \sum_{n\le N} \tilde f_n u^n = \sum_{n\le N_B} e_n u^n + \sum_{N<n\le N_B} \tilde f_n u^n + \sum_{n>N_B} f_n u^n$, and Cauchy's estimate gives $|f_n| \le M\rho^{-n}$. □

For the first-contact and second-contact pieces the computed coefficients are those of f itself ($e = 0$): they are obtained by multiplying and composing power series about $u = 0$, which is exact up to order $N_B = 64$. For the middle pieces, the series of $\mathcal{F}_j$ about $k = 0$ or $y = 0$ are truncated after n_t terms and re-centred exactly, so that $\tilde{f}$ is f with $\mathcal{F}_j$ replaced by its truncation and $e = \Pi \sum_j c_j(\mathcal{F}_j - \tilde{\mathcal{F}}_j)$. If $|e| \le \eta$ on $|u| = \rho$, Cauchy's estimate gives $\eta' = \eta/(1 - w/\rho)$.

S2.6 Proof of Theorem 1

Let the closed disk $|u| \le \rho_2$ satisfy Lemma 2, and let $w < \rho < \rho_2$, with $\rho < 1$ ($\rho < k_0$ and $k_0 + \rho < 1$ for the middle pieces in k; $\rho \le \kappa_0$ and $\kappa_0 + \rho < 1$ for those in κ). Then $P_{\gamma,J}$ and its parts are analytic on $|u| \le \rho$: Π and c_j by Lemma 2, and the Gauss series because their arguments stay in the unit disk. On $|u| = \rho$ we have $|x^{\gamma+1}| \le X^{\gamma+1}$, $|(4pz)^{\gamma+1}| \le (4pZ)^{\gamma+1}$ and $|\sqrt{k}| \le \sqrt{\max|k|}$, and C_j from Equation (S18). With $\Pi^* = X^{\gamma+1}/(2(\gamma+1))$, Equation (S20) applies to the following parts.

First contact. $P_{\gamma,J} = k^m R(k)$ with $R = -(4pz)^{\gamma+1}/(2(\gamma+1)) \sum_{j \le J} c_j H_j$ and $M = (4pZ)^{\gamma+1}/(2(\gamma+1)) \sum_j C_j \hat{H}_j(\rho)$. The error is w^m times the right-hand side of Equation (S20) with $\eta' = 0$.

Middle pieces in k. $f = P_{\gamma,J}$, $M = \Pi^* \sqrt{k_0 + \rho} \sum_j C_j \hat{H}_j(k_0+\rho)$ and $\eta = \Pi^* \sqrt{k_0 + \rho} \sum_j C_j \hat{H}_j^{>n_t}(k_0+\rho)$.

Middle pieces in κ. $f = P_{\gamma,J}$ and $M = \pi\Pi^* \sum_j C_j$ by Lemma 4(c), since $|1-\kappa| \le 1-\kappa_0+\rho \le 1$. With $r_y = 1-\kappa_0+\rho < 1$, $\operatorname{Re} y \ge 1-\kappa_0-\rho > 0$ and $|\ln y| \le \Lambda = \max\big(|\ln(1-\kappa_0-\rho)|, |\ln r_y|\big) + \pi/2$,

$$\eta = \Pi^* \sum_j C_j \left[\hat{A}_j^{>n_t}(r_y) + r_y^{m+j} \big(\Lambda\, \hat{B}_{l,j}^{>n_t}(r_y) + \hat{B}_{c,j}^{>n_t}(r_y) \big) \right]. \tag{S21}$$

Second contact. $P_{\gamma,J} = R + u^m(S_L \ln u + S_C)$ with $R = \Pi \sum_j c_j A_j$, $S_L = \Pi \sum_j c_j u^j B_{l,j}$ and $S_C = \Pi \sum_j c_j u^j B_{c,j}$, and $M_R = \Pi^\dagger \sum_j C_j \hat{A}_j(\rho)$, $M_L = \Pi^\dagger \sum_j C_j \rho^j \hat{B}_{l,j}(\rho)$, $M_C = \Pi^\dagger \sum_j C_j \rho^j \hat{B}_{c,j}(\rho)$, where $\Pi^\dagger = \Pi^* \sqrt{1+\rho}$ outside and Π^* inside. By Lemma 4(d) the error is at most the bound of Equation (S20) for R, plus $L^*(w)$ times that for S_L, plus w^m times that for S_C.

In each case $E_\gamma^{(n)}$ is the resulting bound and $P_{\gamma,J} - \hat{P}_\gamma$ is at most $E_\gamma^{(n)}$; Lemma 1 adds $E_\gamma^{(j)}$. The implementation stores $\sum_i c_i \hat{P}_{\gamma_i}$. For parts shared by all exponents (R and the middle pieces) it uses $\rho = \min_i \rho_i$ and $M = \sum_i |c_i| M_i$, which is valid by the maximum principle; the parts carrying u^{m_i} are bounded separately for each exponent. Dividing $\sum_i c_i P_{\gamma_i}$ by the stellar flux $2\pi \sum_i c_i/(\alpha_i + 2)$ gives the flux, which completes the proof. □

References

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